

Skills Summary

Graphing Proportional Relationships

Use this reference while completing your worksheet or when studying.

The test for proportional. The graph is a *straight line* and it passes through the *origin* (0,0). Both halves are needed: a straight line that misses the origin is linear but not proportional.

The constant of proportionality (unit rate). $k = \frac{y}{x}$ at any point of the line except the origin. Equivalently k is the rise over the run.

The equation. $y = kx$ — no added constant, because the line starts at (0,0). Reading a point off the graph gives k ; the equation then answers questions the drawn axis is too short for.

1. From a table to a graph through the origin

Plot each row as a point (x, y) and rule one straight line through them. If the relationship is proportional the line hits (0,0), so (0,0) is always a legal extra point to plot. The unit rate is then the y -value at $x = 1$.

Example: A table gives 3 hours of work paying 12 dollars. Plot it with the origin and find the unit rate.

Step 1. Pay with no work done is 0 dollars, so (0,0) belongs on the graph; with (3,12) that is enough to rule the line and read its steepness.

Step 2. $k = \frac{12 - 0}{3 - 0} = \frac{12}{3}$. The line through the origin rises 4 dollars for every hour.

Try it: A table gives 4 kilograms of apples costing 10 dollars. What is the cost of one kilogram, in dollars?

check: 2.5

2. Reading the unit rate off a drawn graph

Find a point where the line crosses a corner of the grid, then divide: the y -axis quantity goes on top. Dividing the wrong way round gives the reciprocal — a small number where a large one belongs.

Example: A line through the origin passes through (6,9). Find its unit rate.

Step 1. The line already passes through the origin, so one clean lattice point is all the information a unit rate needs.

Step 2. $k = \frac{9}{6}$, which is the rise 9 over the run 6. The unit rate is 1.5 per unit of x .

Try it: A different line through the origin passes through (8,28). What is its unit rate?

check: 3.5

3. Using the equation to predict

Once k is known, $y = kx$ answers both directions: substitute an x to predict a y , or set y to a target value and solve for x . This is how you answer a question whose numbers run past the end of the drawn axis.

Example: A graph through the origin has unit rate $k = 4$. Predict y when $x = 7$, a value past the right-hand end of the drawn axis.

Step 1. The graph stops before $x = 7$, so read k off the graph and let the equation $y = kx$ carry the prediction.

Step 2. $y = 4 \times 7$. The predicted output is **28**.

Try it: A line through the origin has unit rate 6. Solve $6x = 45$ to find the x that gives an output of 45.

check: 2.5

(!) Watch out: a straight line is not enough. Check the origin before calling a relationship proportional, and check that every $y \div x$ in a table gives the same number. Dividing x by y is the other frequent slip: the unit rate answers “how much y per one x ”, so y is on top.